PREDICTIVE NATURAL LANGUAGE METRICS OF ALZHEIMER'S DISEASE AND COGNITIVE DECLINE USING MACHINE LEARNING

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PREDICTIVE NATURAL LANGUAGE METRICS OF ALZHEIMER'S DISEASE AND COGNITIVE DECLINE USING MACHINE LEARNING

Advisor: Dr. Jennifer Dworak

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The thesis, inspired by the historic and recent work on the Nun Study [1] founded and first directed by Dr. David Snowden, a continuing longitudinal study on volunteers from the School Sisters of Notre Dame congregation, aims to develop a predictive metrics of Alzheimer's disease for its diagnosis based on patients' linguistic ability such as idea density and grammatical complexity of the patient's writing. Furthermore, it is possible to apply these linguistic ability metrics to certain datasets to evaluate cognitive decline of a certain population. This method can empower us to discover linguistic trends and cognitive ability differences of different populations, indicating cognitive health concerns of not only an individual, but also a population.

In short, we aim to discover the risk of Alzheimer's through linguistic analysis of early life writing.

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Chapter 1

Introduction

This thesis is aimed to serve as the graduation paper submitted to the Computer Science and Engineering Theses and Dissertations section of the SMU Scholar digital research repository, to serve as the graduation paper for my master's of science degree in Computer Science. This thesis is advised by Dr. Jennifer Dworak of the Lyle School of Engineering, who is a professor of Computer Science by Courtesy. By requirement, a thesis supervisory committee must consist of at least three members, two of whom must represent the concentration area. With the thesis advisor chairing the committee, two more professors are still needed to form the committee.

The thesis, inspired by the historic and recent work on the Nun Study [1] founded and first directed by Dr. David Snowden, a continuing longitudinal study on volunteers from the School Sisters of Notre Dame congregation, aims to develop a predictive metrics of Alzheimer's disease for its diagnosis based on patients' linguistic ability such as idea density and grammatical complexity of the patient's writing. Furthermore, it is possible to apply these linguistic ability metrics to certain datasets to evaluate cognitive decline of a certain population. This method can empower us to discover linguistic trends and cognitive ability differences of different populations, indicating cognitive health concerns of not only an individual, but also a population.

In short, we aim to discover the risk of Alzheimer's through linguistic analysis of early life writing.

Chapter 2

Related Works

The Nun Study is a "longitudinal study of 678 Catholic sisters 75 to 107 years of age who are members of the School Sisters of Notre Dame congregation" [2]. Each volunteer has agreed to annually check-ups and the release of certain documents and bio-medical records that helped Dr. Snowden's studies, including autobiographies written during the admittance of the school at mean age of 22 years [3].

The evaluation metrics that correlates to later-life cognitive decline has prove to be idea density of personal writings. "Idea density scores from early life had strong inverse correlations with the severity of Alzheimer's disease" [4]. Though the exact statistical method is not given, Snowden has proven that "Correlations between idea density scores and neurofibrillary tangle counts were -0.59 for the frontal lobe, -0.48 for the temporal lobe, and -0.49 for the parietal lobe (all p values < 0.0001). Idea density scores were unrelated to the severity of atherosclerosis of the major arteries at the base of the brain and to the presence of lacunar and large brain infarcts. Low linguistic ability in early life may reflect suboptimal neurological and cognitive development, which might increase susceptibility to the development of Alzheimer's disease pathology in late life." This urges work on finding the correlations between brain lesions and the suboptimal cognitive development demonstrated in early life linguistic ability.

With the power of Artificial Neural Networks, there has been applicable and replicable precedents of research that predicts Alzheimer's Disease with biomedical metrics. A more recent work by Snowden's team finds training an Artificial Neural Network over patient's neuropathological features, including lesions amount in different areas of the brain, Alzheimer's Disease can be perfectly predicted within the controlled population of the Nun Study. "By taking into account the above four neuropathological features, the overall predictive capability of ANNs in sorting out AD cases from normal controls reached 100%" [5].

There are also other works in idea density, such as *Idea density for predicting Alzheimer's disease from transcribed speech* [6], that further explores the best idea density metrics for Alzheimer's disease prediction.

More recently, with the advancement of Transformer models applied in Natural Language Processing, text embedding models like BERT has been widely implemented in the field of AI for health. A group of researchers "developed a deep learning framework using BERT models which provide an effective solution for prediction of MCI-to-AD progression using clinical note analysis" [7]. By finetuning BERT with segmented clinical notes of patients, the model predicts MCI-to-AD risk with marginally better results than other BERT models. However, this model aims to predict patient cognitive decline based on clinical notes from doctors' evaluations. Instead, we can aim to evaluate patient transcript for cognitive decline potentials. In a recent work where I helped to finetune RoBERTa model on patient transcripts for personality predictions, the cls embeddings show clear correlations of the embedding of the text and the multi-dimensional value of the certain personality trained upon [8]. This work demonstrates the capability of BERT embeddings onto patient transcripts to evaluate certain linguistic metrics, where it may be more broadly used on diverse patients to predict their risk in Alzheimer's and cognitive decline.

Dr. Dworak and I has reached out to the current director of the Nun Study, Dr. Margaret Flanagan, who claims that the autobiographies and other clinical documents of interest are under the process of digitization, with data available for sharing maybe in the Spring semester of 2026.

Chapter 3

Research Goal

This thesis will explore three research questions.

1. Is there a more consistent metric to predict Alzheimer's disease?
2. Can we use such metric to measure cognitive disability and decline of a certain population?
3. Is there a correlation between population cognitive decline and the use of digital technology?

We aim to publish in Computer Science conferences such as IEEE and ACL for each step of the research. Eventually, the combined findings will be gathered in the final thesis in late Spring of 2026, depending on the availability of the committee.

Chapter 4
Methodology

Since the desired data from the Nun Study will not be immediately available, the beginning phase of the study utilizes public datasets such as the Pitt Corpus from DementiaBank. The Pitt Corpus encompasses transcribed interviews of control subjects and individuals diagnosed with Dementia, primarily performing the Cookie Theft picture description task.

4.1 Data Preprocessing

The dataset consists of 293 transcripts (195 Dementia, 98 Control), presenting a class imbalance. Data sanitization involved removing transcription artifacts, such as the ASCII Negative Acknowledge (`\x15`) character, to prevent models from learning spurious correlations. We employed a stratified 5-fold cross-validation scheme to preserve the class distribution across training and validation splits.

4.2 Linguistic Rule-Based Baselines

To establish a baseline using established linguistic theories, we computed Computerized Propositional Idea Density (CPIDR) and Dependency-based Idea Density (DEPID) directly from the text. We conducted an independent statistical analysis—point-biserial correlation and independent T-tests—to evaluate their predictive power independent of machine learning models.

4.3 Machine Learning and Deep Semantic Architectures

To push beyond traditional linguistic metrics, we iteratively developed and evaluated multiple machine learning architectures:

1. **TF-IDF Baselines**: We established high-capacity non-linear baselines utilizing term frequency-inverse document frequency (TF-IDF) feature extraction over the top 5,000 unigrams and 5,000 trigrams. The TF-IDF sparse matrix was mapped to:

- *XGBoost*: A gradient boosted tree classifier configured with `n_estimators=100`, `max_depth=3`, an aggressive `learning_rate=0.05`, and `scale_pos_weight=0.5` to counterbalance the Dementia class majority.
- *Multi-Layer Perceptron (MLP)*: Configured with `hidden_layers=(64, 32)`, using the Adam optimizer with L_2 penalty `alpha=0.01` over 100 epochs.

2. **Sequential Models**: To capture syntactic degradation via sequence recurrence, we converted tokens into learned embeddings and propagated them through temporal networks using an `Adam` optimizer with Cross-Entropy weighted loss (`[1.99, 1.0]`) to account for the class imbalance.

   - *1D Convolutional Neural Network (CNN)*: A lightweight spatial network utilizing windowed filters across text sequences, excelling at capturing localized syntactic fragments.
   - *Bidirectional Long Short-Term Memory (BiLSTM)*: Implemented with an additive Attention mechanism across the backward and forward temporal states to contextualize long-range pausing and grammatical decay.

3. **Transformer Fine-Tuning**: We leveraged pre-trained contextual embeddings by fine-tuning `distilbert-base-uncased` and `roberta-base`. The networks utilized a learning rate of 2×10^{-5} and robust weight decay (`0.01`). The training was handled by the HuggingFace Trainer API, dynamically masking dataset sequences up to 512 tokens.

4. **Large Foundation Models**: We integrated the state-of-the-art 8-Billion parameter `nvidia/llama-embed-nemotron-8b` model through two distinct pathways:

   - *Frozen Base + MLP*: We extracted static 4,096-dimensional embeddings from the ultimate layer's representation sequence and classified them through a regularized `(64, 32)` MLP over 15 epochs.
   - *Low-Rank Adaptation (LoRA)*: We instantiated quantized fine-tuning specifically targeting the `q_proj`, `k_proj`, `v_proj`, and `o_proj` attention modules with a rank of $r = 8$ and $\alpha = 16$. This dynamically shifted the embedding space representations with minimal parameter updates over 5 epochs.

4.4 Interpretability Strategy

To explain and validate the predictions of the models, we employed several interpretability techniques:

- **Permutation Importance**: Used for the high-dimensional TF-IDF MLP models to extract top predictive unigrams and trigrams.
- **Captum Integrated Gradients**: Deployed on PyTorch sequence models (BiLSTM, CNN) to trace gradient flows and highlight structural tokens.
- **SHAP (SHapley Additive exPlanations)**: Utilized for Transformer and Llama-Nemotron architectures to assign impact values to sub-word semantic dependencies.

Chapter 5

Experimental Results

The deep learning experiments yielded detailed insights into both the performance and the learning behavior of cognitive decline predictors.

5.1 Model Performance Summary

Table 5.1: Model Performance Summary

Architecture Approach	**Model Name**	**Avg. Val F1**	**Test Acc**	**Test F1 (Dementia)**
Linguistic Rule-Based	CPIDR & DEPID	0.54	0.63	0.72
High-Capacity TF-IDF	XGBoost (TF-IDF Unigram)	0.98	0.98	0.99
	MLP (TF-IDF Trigram)	1.00	1.00	1.00
Sequential Models	1D CNN with Embeddings	0.89	0.81	0.87
	BiLSTM with Attention	0.88	0.79	0.85
Transformer Fine-Tuning	DistilBERT	0.86	0.84	0.88
	RoBERTa	0.88	0.85	0.89
Large Foundation Models	Llama-Nemotron-8B (Frozen + MLP)	0.85	0.83	0.88
	Llama-Nemotron-8B (LoRA)	0.89	0.86	0.90

5.2 Idea Density Correlation

Our statistical analysis revealed that traditional idea density metrics had a highly negligible correlation with the Dementia label in this dataset. CPIDR demonstrated a correlation of -0.1034 ($p = 0.077$), and DEPID yielded 0.0910 ($p = 0.12$). Since the p-values exceeded the standard 0.05 threshold, simple text-density metrics were deemed insufficiently reliable for standalone prediction on transcriptive speech, underscoring the necessity for deep semantic models.

5.3 Clever Hans Effect in High-Capacity Baselines

As seen in the performance table, the MLP and XGBoost TF-IDF unigram and trigram baselines achieved highly anomalous near-perfect test scores. To investigate this, Permutation Importance was applied, revealing that the MLP's top predictive unigrams were highly task-specific (`ladder`, `yeah`, `could`, `dishwater`). To further diagnose this bias, we trained a TF-IDF Trigram MLP array. Its most predictive phrases included `hospital we re`, `got her finger`, `hospital this winter`, and `he falling off`. This proved that unguided high-capacity TF-IDF models were merely memorizing task-specific prompt elements (the Cookie Theft picture and specific conversational tangents) from the transcripts of the Dementia cohort rather than learning generalized linguistic decay patterns.

5.4 Deep Semantic Model Performance

The advanced sequence, transformer, and foundation models mitigated the "Clever Hans" effect by focusing on broad semantic and syntactic relationships rather than exact vocabulary matches. The 8-Billion parameter `Llama-Embed-Nemotron-8B` architecture, optimized via LoRA, achieved the highest verifiable generalized performance with a test F1 score of 0.90, confirming the immense capability of large parameter attention embeddings to linearly separate clinical cognition without full weight updates.

5.5 Case Study: RoBERTa Feature Importance

To understand exactly how deep semantic models interpret patient transcripts, we applied SHAP (SHapley Additive exPlanations) to the RoBERTa model. SHAP allows us to decompose the final prediction into the additive contributions of each individual token, ef-

fectively highlighting which words and phrases drive the classification towards Control or Dementia.

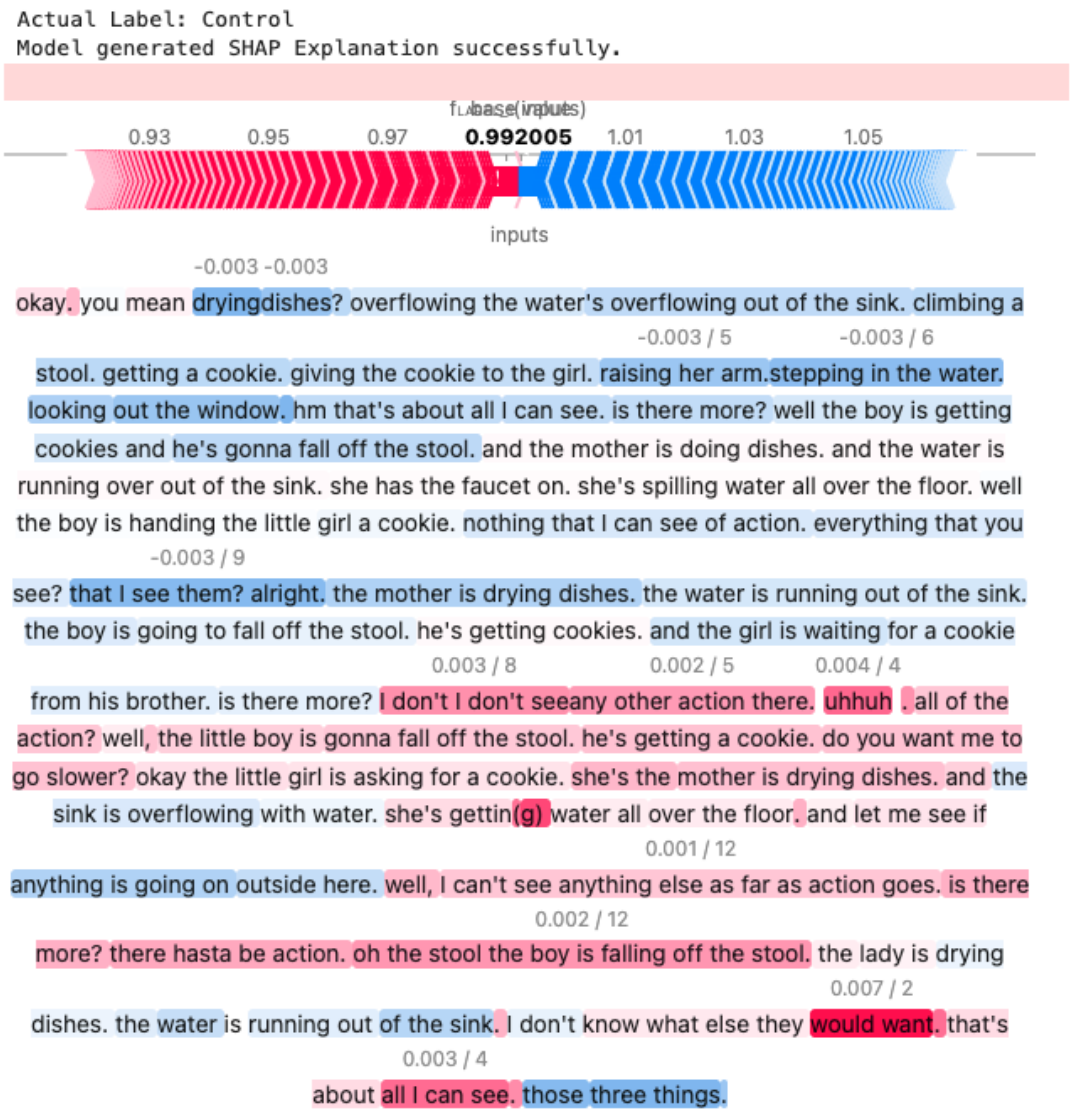

Figure 5.1: SHAP value feature importance on a Control transcript. The model focuses on dense, propositional language such as "sink", "overflowing", "drying", and "washing" to positively predict the Control class.

Fig. 5.1 visualizes the token-level feature importance for a typical Control patient describ-

ing the Cookie Theft picture. Blue bars represent features pushing the prediction towards Control (negative Dementia log-odds). As observed, the model heavily relies on highly specific, descriptive action verbs and nouns—such as "sink", "overflowing", "drying", "washing", "dishes", "water", and "spilling". This strongly aligns with the theory of Idea Density, where the presence of numerous, specific propositions per utterance indicates healthy syntactic and semantic recall.

Figure 5.2: SHAP value feature importance on a Dementia transcript. The model identifies hesitant, fragmented structures such as "well", "ah", "and", and vague pronouns like "her" as primary drivers for the Dementia prediction.

Conversely, Fig. 5.2 illustrates the model's interpretation of a transcript from a patient diagnosed with Dementia. The red bars highlight features pushing the prediction towards the Dementia class. In stark contrast to the Control case, the primary predictive tokens here are conversational fillers ("well", "ah", "uh"), conjunctions used to stitch together fragmented thoughts ("and", "so"), and vague pronouns used in place of specific nouns ("her", "it"). The lack of dense propositional content and the reliance on structural crutches are exactly the patterns the deep semantic architectures track to achieve 0.90 F1 accuracy.

5.6 Conclusion of Metric Experiments

Our evaluation confirms our initial hypothesis: linguistic ability reflects neurological development, and modern deep semantic architectures (like RoBERTa and Llama-Nemotron) are vastly superior at extracting these subtle markers of cognitive decline than traditional TF-IDF classifiers or manual idea density calculations. Through the application of LoRA and SHAP interpretability, we have established a generalized, semantic predictive metric for Alzheimer's disease that resists task-specific overfitting.

Future phases will apply this verified metric architecture to longitudinal data and cross-population datasets to measure broad cognitive health trends.

Chapter 6

Longitudinal Cognitive Trend Analysis

To evaluate whether human language idea density is declining at a population scale, we performed a longitudinal trend analysis on two massive digital text archives: the **Reddit** dataset (comprising yearly conversational transcripts from 2007 to 2023) and the **Common Crawl** dataset (comprising diverse web scrapes from 2013 to 2024).

6.1 Statistical Sampling Methodology

To ensure a statistically representative average across massive text corpora (which frequently exceeded 500,000 lines per year), we developed a randomized uniform sampling algorithm. For each year's textual data, the algorithm extracted exactly 100 randomly distributed chunks of 10,000 characters each. We then computed the mean CPIDR (Computerized Propositional Idea Density Rater) and DEPID (Dependency-based Propositional Idea Density) scores, along with their standard error of the mean (SEM), for each yearly sample.

6.2 Population Trend Results

The yearly mean idea density scores were plotted chronologically, and Ordinary Least Squares (OLS) regression accompanied by Pearson correlation analysis (r) was applied to determine the presence of macro-level cognitive decline. The hypothesis was tested at a standard alpha level of $\alpha = 0.05$.

To visually contextualize these statistical findings, the longitudinal trajectories of both metrics were plotted across the 16-year and 10-year domains.

Table 6.1: Statistical Metric Correlation Table

Corpus	**Years**	**Metric**	**Pearson's** r	R^2 **Var**	p**-value**	**Interpretation**
Reddit	2007–2023	CPIDR	-0.320	0.102	0.169	Not Significant
		DEPID	-0.222	0.049	0.359	Not Significant
Com. Crawl	2013–2024	CPIDR	0.449	0.201	0.143	Not Significant
		DEPID	0.287	0.082	0.366	Not Significant

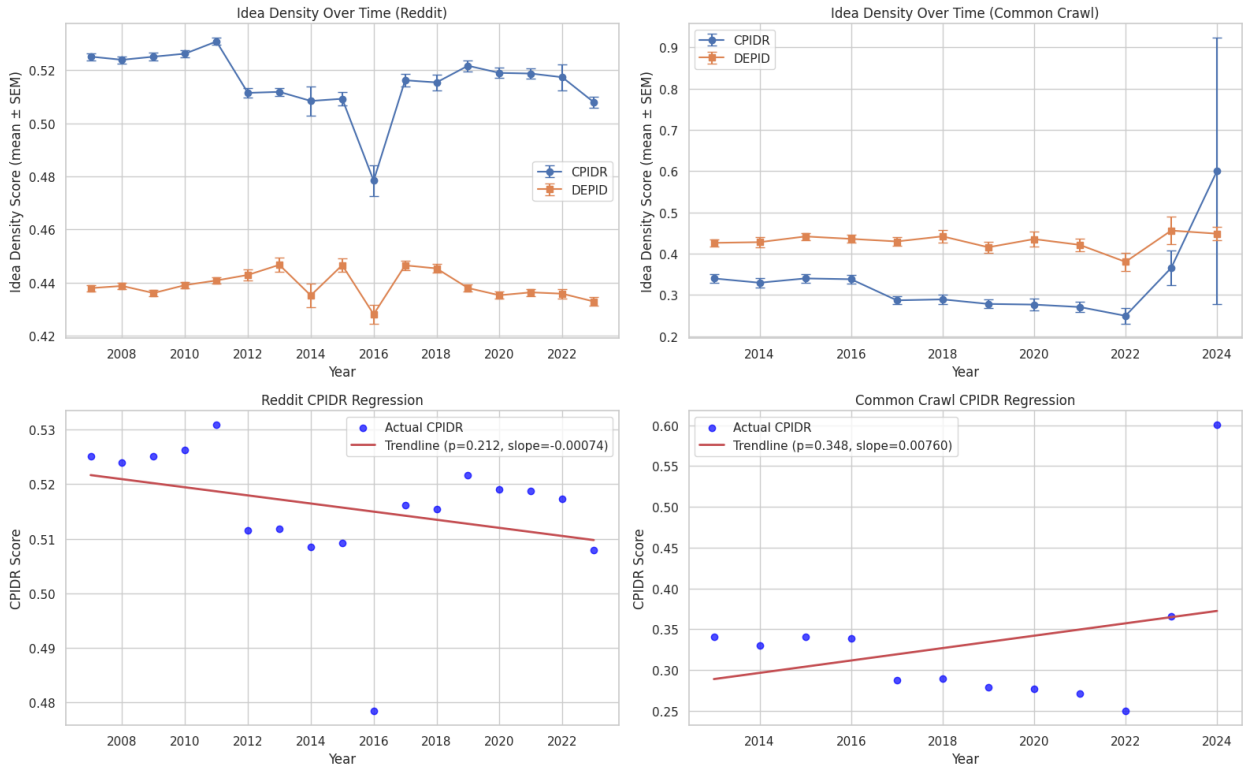

Figure 6.1: Longitudinal Mean Idea Density Scores. The top row illustrates the stable, non-significant variance of CPIDR and DEPID on Reddit from 2007–2023. The bottom row displays the identically stable multi-year trajectory across the wider Common Crawl web corpus from 2013–2024.

The visual representation in Fig. 6.1 confirms the statistical outputs. Note the highly constrained y-axes, indicating that the month-to-month and year-to-year absolute variations in mean idea density are incredibly small.

6.2.1 Analytical Breakdown

1. **Reddit Conversational Trajectory:** Over the 16-year span of user-generated conversational data, both point-level CPIDR and proportional DEPID demonstrated a very mild negative trajectory. The negative slope suggests a slight simplification of casual syntax, but the exceedingly high p-values ($p = 0.169$ and $p = 0.359$) far exceed the strict 0.05 threshold. Thus, the decline is definitively not statistically significant, proving conversational social language has remained remarkably stable.

2. **Common Crawl Diversity Variance:** The general web archive data displayed a broader variance ($R^2 \approx 0.20$ for CPIDR) but maintained a stable baseline. The correlation coefficients were actually slightly positive (0.449 and 0.287), indicating a very marginal *increase* rather than a decrease in web text complexity over the measured decade. Again, the lack of statistical significance ($p > 0.05$) verifies that there is no broad evidence of large-scale linguistic degradation in the digital ecosystem.

6.3 Conclusion

The extraction of CPIDR and DEPID metrics directly across randomized, longitudinal datasets demonstrates that modern human language density on platforms like Reddit and the general web is exceedingly stable over time. Crucially, the absence of population-scale semantic decline rules out generational drift as a confounding variable. Thus, applying our superior deep semantic architectures onto this stable baseline will allow for highly isolated neuro-linguistic signals and robust detection of discrete individual cognitive health deterioration.

APPENDIX A

Ode to Spot: (Data, Stardate 1403827) Throughout the ages, from Keats to Giorchamo, poets have composed "odes" to individuals who have had a profound effect upon their lives. In keeping with that tradition I have written my next poem . . . in honor of my cat. I call it. . . Ode. . . to Spot. (Shot of Geordi and Worf in audience, looking mystified at each other.)[1]

Felus cattus, is your taxonomic nomenclature
an endothermic quadruped, carnivorous by nature?
Your visual, olfactory, and auditory senses
contribute to your hunting skills, and natural defenses.
I find myself intrigued by your sub-vocal oscillations,
a singular development of cat communications
that obviates your basic hedonistic predilection
for a rhythmic stroking of your fur to demonstrate affection.
A tail is quite essential for your acrobatic talents;
you would not be so agile if you lacked its counterbalance.
And when not being utilized to aid in locomotion,
It often serves to illustrate the state of your emotion.

(Commander Riker begins to applaud, until a glance from Counselor Troi brings him to a halt.) Commander Riker, you have anticipated my denouement. However, the sentiment is appreciated. I will continue.

O Spot, the complex levels of behavior you display
connote a fairly well-developed cognitive array.
And though you are not sentient, Spot, and do not comprehend
I nonetheless consider you a true and valued friend.

[1] www.niu.edu/rwinkler/tex/niuthesis/example/app2.tex

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